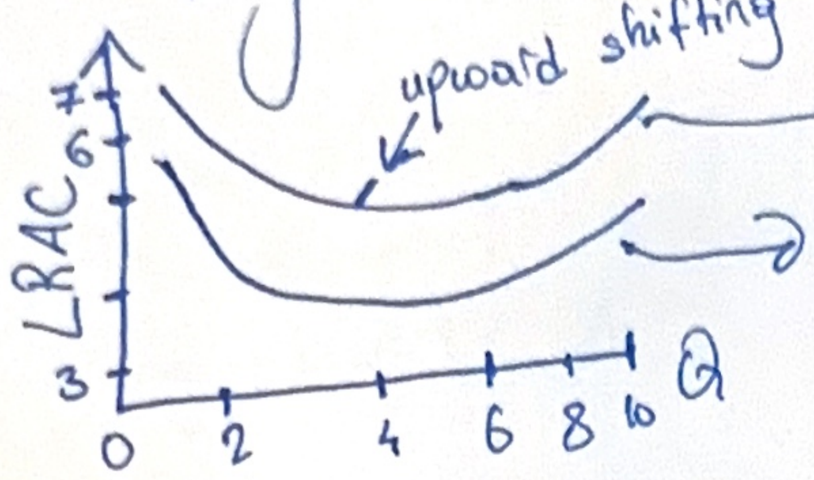
HW6 - Cost Analyses.

P 8.2

A) A rise in wage rates.



New LRAC (higher wages)

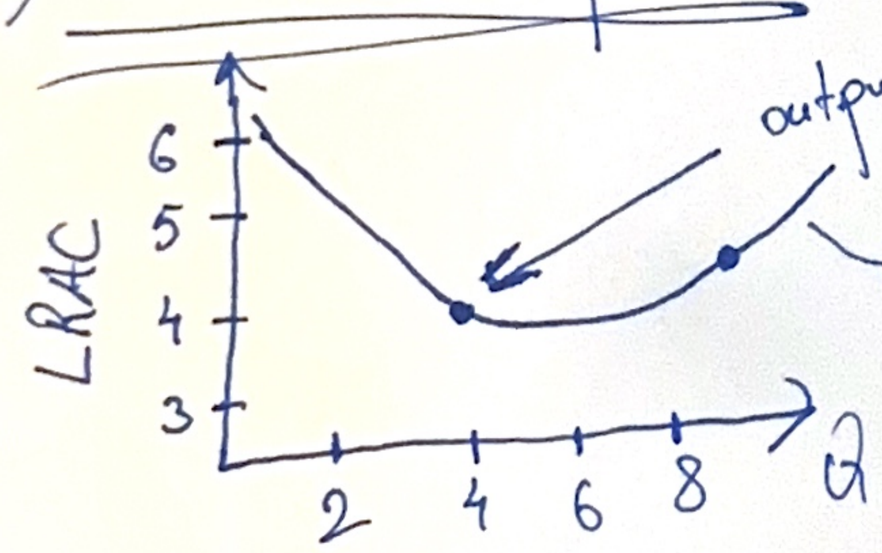
Original LRAC

because Wages are a production input cost.

Higher wages increase cost of Labor.

Effect on AC increasing

B) Decline in output



AC may rise or fall depending on starting point

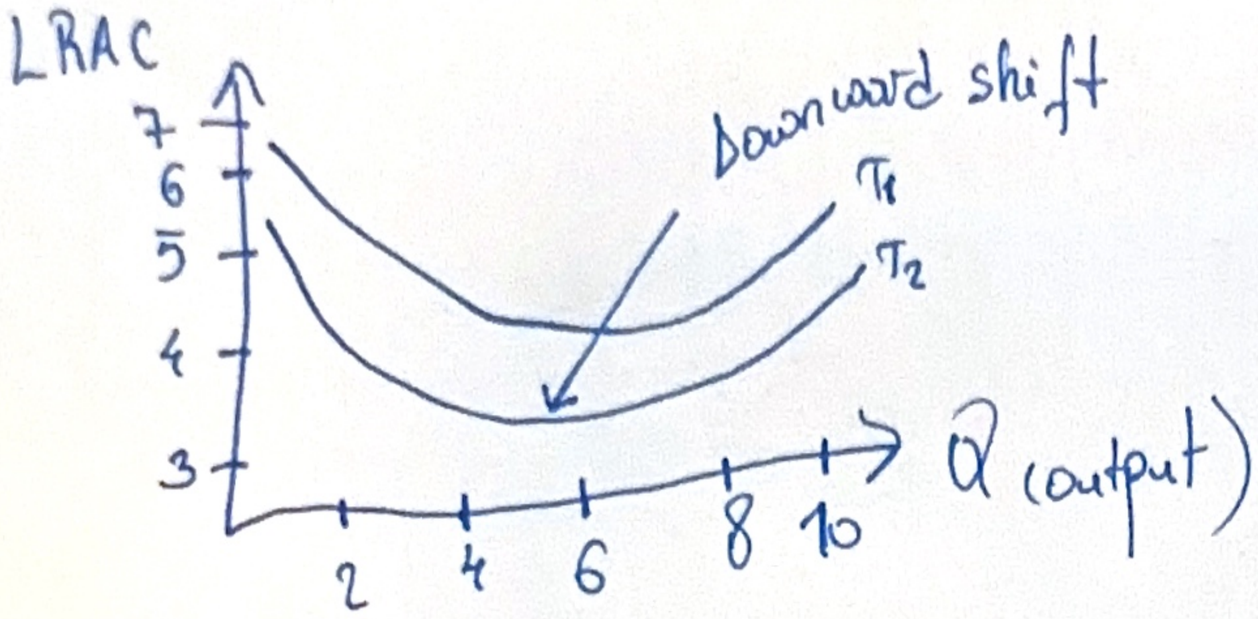
Leftward movement along LRAC, AC is uncertain.

Output decrease, so firm moves to a lower quantity on the existing LRAC curve.

The curve itself does not shift Because there is no change in input prices or technology.

C) An Energy - saving technical change

It's downward shift in LRAC, AC decreases.



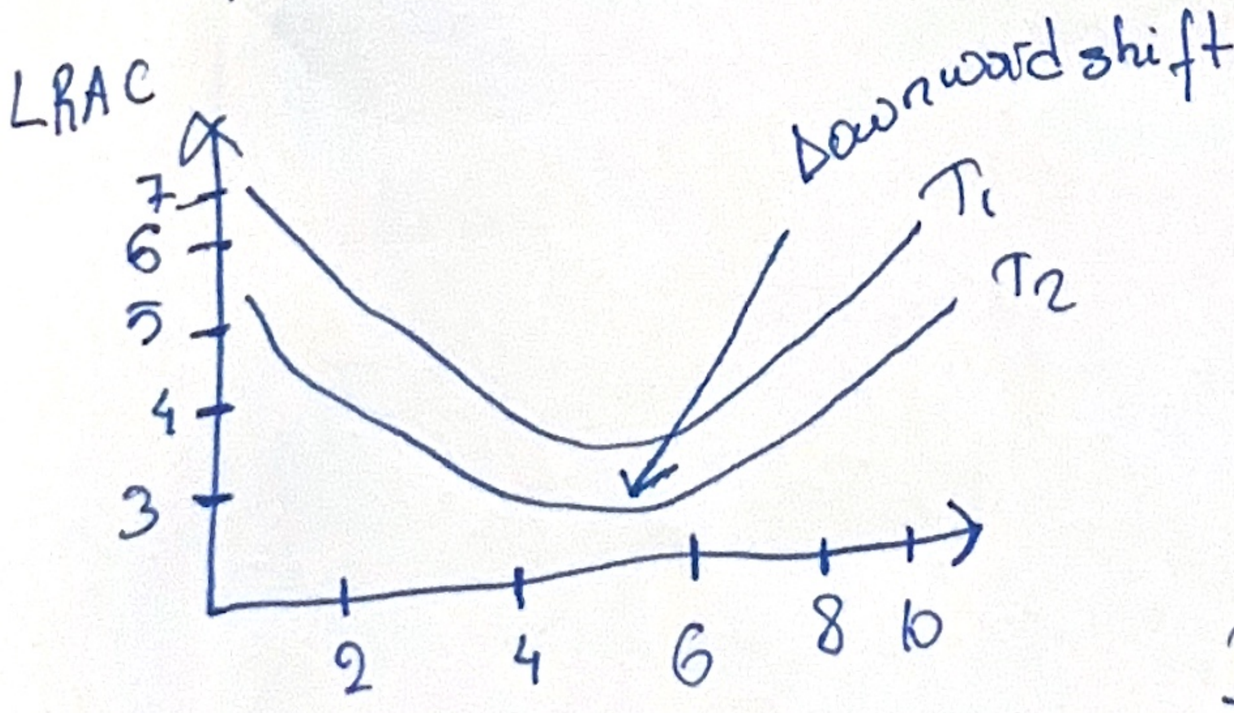
T_1 : Original LRAC

T_2 : New LRAC (lower cost)

Energy - saving ⇉ reduces amount of energy required or cost of per unit of output.

Production becomes cheaper at every output level

D) A fall in interest rates.



T_1 : Original LRAC

T_2 : New LRAC (lower cost)

AC : decreasing

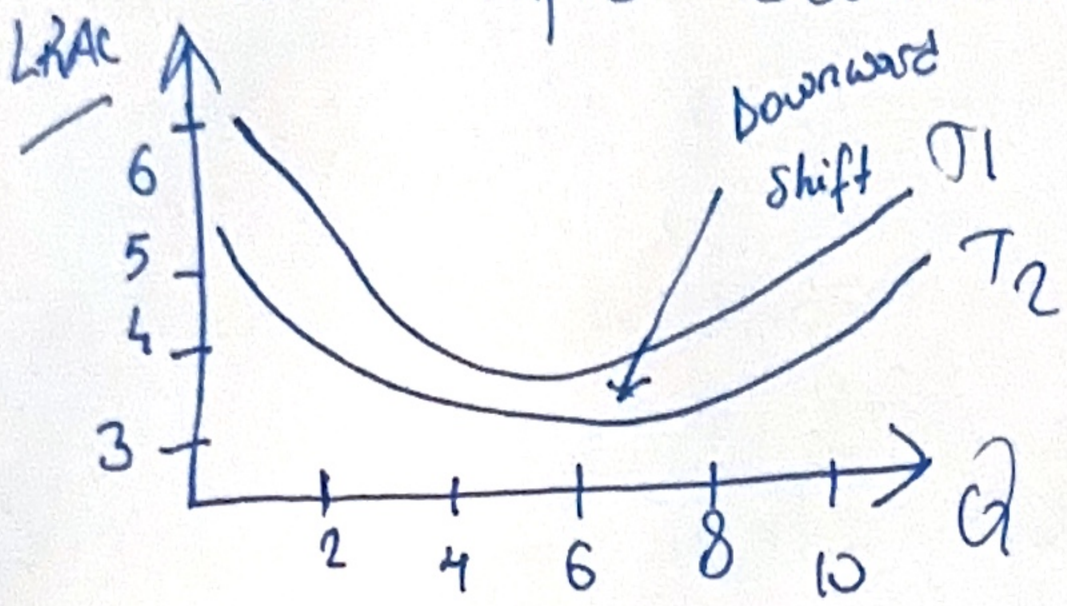
Interest rates reps the cost of borrowing capital.

Lower interest rates reduces the cost of financing machinery

Firm's LR production cost decrease.

E) An increase in learning or experience.

LRAC shifts. downward.



T_1: original LRAC

T_2: New LRAC (lower cost)

AC: Decreasing

Workers become more skilled & efficient.

Production process improve.

Less time, Labor, & resources are required to produce same output.

P 8.4 Accounting & Economic cost

Given

of students ⇒ 3

Summer salary per student ⇒ 8,000 $

selling price per smoothie = 5 $

Facility lease = 8,000 $

Insurance = 1,000 $

supplies per smoothie = 3.2 $

Q = number of smoothie sold in the summer

P = 5 $

TC = total cost

TR = Total revenue

* Accounting cost
* Economic cost

A) Accounting cost function (Explicit costs)

$$FC = 8,000 + 1,000 = 9,000\ \$$$

$$VC = 3.2Q$$

$$TC_A = FC + VC$$

$$TC_A(Q) = 9,000 + 3.20Q$$

B) Economic cost function

Opportunity cost = $3 \times (6{,}000) = 18{,}000$ ₪

$FC_E = 8{,}000 + 1{,}000 + 18{,}000$

$FC_E = 27{,}000$ ₪ $\qquad VC = 3.2Q$

$TC_E = FC_E + VC$

$= 27{,}000 + 3.2Q$

Economic cost $= 27{,}000 + 3.2Q$

C) Economic break-even number of units.

↳ business earns enough to cover both

Profit = TR − TC

= 0

TR = Total Economic cost.

$5 \times 15{,}000 = 75{,}000$

$TR = P \times Q$ → $TC_E = 27{,}000 + 3.2(15{,}000) = 75{,}000$

$TR = 5Q$ →

$TC_E = 27{,}000 + 3.2Q$

Economic break-even point is 15,000 units.

$TR = TC_E$

$5Q - 3.2Q = 27{,}000$

$Q = \frac{27{,}000}{1.8} =$ 15,000 smoothies

P.87

Given

AFC = 4,500 /month $\Rightarrow$ $AFC = \frac{TFC}{Q}$

$AVC = 59 - 0.006Q$ $\rightarrow$ $AVC = \frac{TVC}{Q}$

commission per trade = $100

Sales rep commission = 35%

Target monthly return = 7,500 $

$TC = TFC + TVC$

$Profit = TR - TC$

A) solution

$TFC = 4.500$ $

$AVC = 59 - 0.006Q$ $\overset{Q}{=} Q * AVC = \frac{TVC}{Q} \times Q$

$TVC = (59 - 0.006Q)Q$

$TVC = 59. - 0.006Q^2$

$TC = 4,500 + 59 - 0.006Q^2$

sales commission = $100 \times (0.35) = 35$ $

Net commission = $100 - 35 = 65$ $

$TR = 65Q$

$Profit = 65Q - (4,500 + 59Q - 0.006Q^2)$ Profit = 7,500

$= 0.006Q^2 + 6Q - 4,500 = 7,500$

$0.006Q^2 + 6Q - 12,000 = 0$ $Q_1 = 1,000$

$Q_2 = -2,000$

B P.8.7

$Q = 1000$

$$TC = 4{,}500 + 59Q - 0.006Q^2$$

$$E_C = \frac{dTC}{dQ} \times \frac{Q}{TC}$$

$$E_C = \frac{MC}{AC}$$

$$MC = \frac{dTC}{dQ} = 59 - 0.012Q \qquad Q = 1000$$

$$MC = 59 - (0.12 \times 1000) = 47\ \$$$

$$TC = 4{,}500 + 59Q - 0.006Q^2$$

$$TC = 4{,}500 + 59(1000) - 0.006(1000)^2$$

$$TC = 4{,}500 + 59{,}000 - 6{,}000$$

$$TC = 57{,}500\ \$$$

$$AC = \frac{TC}{Q} = \frac{57{,}500}{1{,}000} = 57.5\ \$$$

$$E_C = \frac{47}{57{,}5} = 0.8174 < 1$$ Economies of scale; AC decrease